

Estimating METs for the Spanish IMPaCT Cohort Using WearablePerMed Dataset

Authors.

Abstract— Abstract

Index Terms— Accelerometers, Energy Expenditure, Inertial Measurements Unit (IMU), Metabolic Equivalent (MET), Neural Networks.

I. INTRODUCTION

Physical activity is a primary behavioral determinant of chronic disease, functional decline, and mortality. Objective quantification of physical behavior at population scale has become feasible through wearable inertial measurement units (IMUs), whose low cost and unobtrusiveness enable continuous monitoring in free-living conditions. Large-scale cohort initiatives—including the UK Biobank, NHANES, and the Spanish IMPaCT study—have adopted IMU-based protocols to complement genetic and clinical data with device-derived behavioral phenotypes, substantially reducing the biases inherent to self-report.

A central methodological decision in cohort-scale IMU deployment concerns sensor placement and configuration. Single wrist-worn devices offer high compliance and operational simplicity, whereas thigh placement provides more reliable estimates of posture and sedentary time. More recently, dual-sensor designs combining wrist and thigh IMUs have been adopted to leverage the complementary strengths of each site. The IMPaCT cohort exemplifies this paradigm: it systematically deploys triaxial accelerometers and gyroscopes simultaneously at both locations in a nationwide sample targeting 200,000 participants, reflecting an explicit commitment to richer behavioral phenotyping in support of Precision Medicine.

Much of the existing literature on IMU-based physical activity analysis frames the problem as discrete activity classification, assigning windows of sensor data to predefined categories such as walking, sitting, or cycling. While intuitive, this formulation carries inherent limitations: it requires exhaustive, mutually exclusive class definitions; annotation is costly and poorly transferable across populations; and it is sensitive to cultural and contextual variability in behavioral repertoires. These constraints are particularly acute in large-

scale heterogeneous cohorts where free-living behavior resists neat categorical boundaries.

Continuous estimation of physical activity intensity expressed as metabolic equivalents (METs) offers a compelling alternative. METs quantify energy expenditure relative to resting metabolic rate, providing a physiologically grounded, class-agnostic scale that is inherently compatible with the behavioral heterogeneity of free-living conditions. MET values are extensively used in clinical guidelines and epidemiological research, and their estimation from IMU signals sidesteps the need for activity label annotation entirely. Doubly labeled water and portable indirect calorimetry constitute the gold standard for MET measurement, but their complexity and cost confine them to controlled protocols; scalable IMU-based estimation is therefore of direct practical relevance for population cohorts. Despite growing interest, systematic evaluation of MET estimation from raw inertial signals remains limited, particularly for dual-sensor configurations representative of contemporary cohort designs. In this work, we address this gap using data from the WearablePerMed project, a dataset specifically designed to replicate the IMPaCT measurement protocol. Structured activities were recorded with simultaneous indirect calorimetry to provide physiological ground truth; additional semi-structured activities were annotated using Compendium of Physical Activities references to extend ecological coverage. We benchmark multiple modeling strategies across wrist-only, thigh-only, and combined wrist-and-thigh configurations, with the central objective of determining whether single-sensor setups can achieve performance comparable to dual-sensor designs—an answer with direct implications for the cost-efficiency of future population-based monitoring. To systematically contextualize the evaluated approaches, a structured literature search was conducted across IEEE Xplore, Scopus, and Google Scholar (2005–2025) to identify studies estimating physical activity intensity or energy expenditure from wearable inertial data; findings informed both the selection of modeling strategies and the framing of the Related Work section.

II. Related Work

To identify studies addressing the estimation of physical activity intensity and energy expenditure from wearable inertial

sensor data, a structured literature search was conducted across IEEE Xplore, Scopus, and Google Scholar, covering publications from 2005 to 2025. Search queries combined terms related to inertial sensing and metabolic outcomes, including accelerometer, IMU, metabolic equivalent (MET), energy expenditure, raw acceleration, machine learning, and deep learning, connected through Boolean operators. Studies were included if they (i) reported quantitative performance results, (ii) involved human participants, (iii) employed wearable inertial sensors to estimate physical activity intensity or energy expenditure as a continuous outcome, and (iv) were published in peer-reviewed journals or conference proceedings. Works were excluded if they addressed activity classification without any intensity estimation component, lacked experimental validation, or were not peer-reviewed.

The initial search retrieved 300 records from database searching, supplemented by 5 additional records identified through other sources, yielding 305 records after duplicate removal. Following title and abstract screening, 275 records were excluded and 30 proceeded to full-text assessment. Of these, 10 were excluded due to the absence of quantitative results, leaving 20 studies in the qualitative synthesis and 8 in the quantitative synthesis. The complete selection process is illustrated in Fig. 1.

Analysis of the selected literature reveals three dominant methodological paradigms. Early work relied on linear or polynomial regression models derived from summary statistics of accelerometer signals — vector magnitude, activity counts, or ENMO — achieving coefficients of determination often exceeding 0.7 under controlled conditions [X][X][X][X]. Despite their interpretability, these models exhibit limited capacity to capture nonlinear relationships between movement dynamics and metabolic cost, restricting their generalizability across diverse activities and populations. Subsequent studies introduced machine learning techniques operating on engineered feature sets, including time-domain statistics, frequency-domain descriptors, autocorrelation measures, and inter-axis correlations [X][X][X]. Ensemble methods such as Random Forests and gradient boosting demonstrated improved prediction accuracy, particularly in multi-activity scenarios, while retaining moderate interpretability. More recently, deep learning architectures — CNNs, LSTMs, and hybrid models — have been applied directly to raw inertial time series, showing enhanced ability to learn activity-agnostic representations of movement intensity and to capture temporal dependencies inherent in human motion [X][X][X]. Nevertheless, systematic evaluation across sensor configurations and the added value of dual-sensor setups for continuous MET estimation remain limited, motivating the present work.

A schematic overview of the study selection process is provided in Fig. 1.

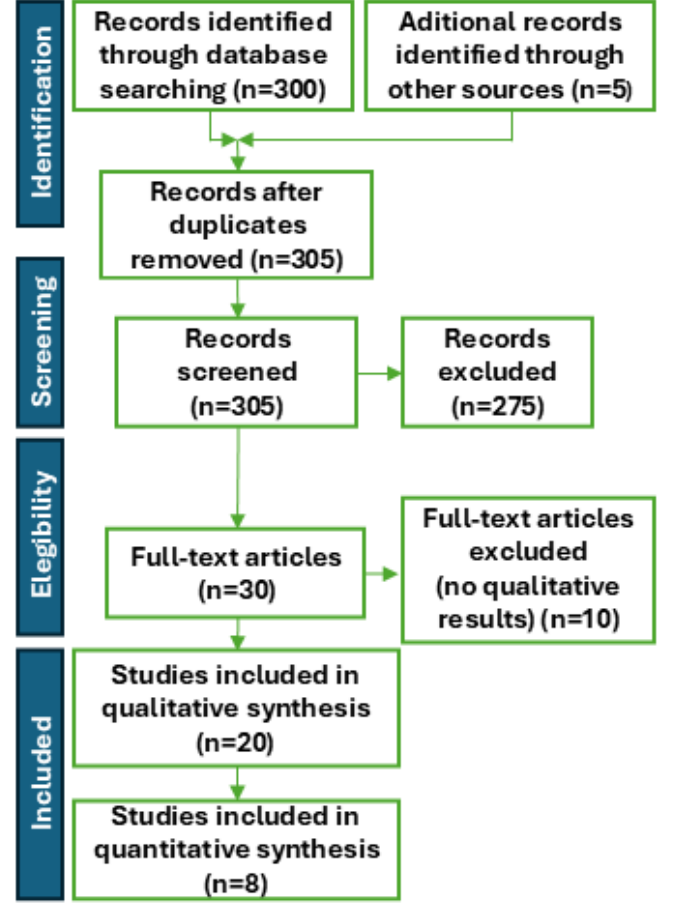


Fig. 1. Flow Diagram of the study according to PRISMA methodology.

IV. Materials And Methods

A. WearablePerMed Dataset

The present study leverages data collected within the WearablePerMed project (PMP22/00028), a national collaborative initiative designed to support methodological research in wearable-based physical activity assessment under cohort-representative conditions. The project was conceived to align with the measurement philosophy of the Spanish IMPaCT cohort, particularly with respect to multimodal inertial sensing and scalable experimental protocols.

The dataset comprises data acquired from a sample of XX adult participants using wearable inertial measurement units (MATRIX IMUs) simultaneously deployed at the wrist and thigh. Each IMU integrates a triaxial accelerometer and a triaxial gyroscope, enabling the capture of both linear acceleration and angular velocity signals. All sensors were configured to record data continuously at a sampling frequency of 25 Hz over a one-week period, encompassing both laboratory-based activity protocols and unsupervised free-living behavior.

The experimental design includes a set of structured laboratory

activities for which energy expenditure was directly measured using a portable indirect calorimetry system (COSMED K5). These activities were selected to cover a range of physical effort levels and controlled movement patterns and included a resting phase, sit-to-stand transitions, treadmill exercise, and an incremental cycle ergometer test. The resting phase lasted 30 minutes, while the sit-to-stand task was performed for 30 seconds, during which participants were instructed to complete as many transitions as possible. The treadmill protocol consisted of seven two-minute bouts increased at 2m/s each resulting in walking or running depending on individual capacity. The cycle ergometer test began with a two-minute resting period followed by successive two-minute stages of increasing workload, terminating at subject exhaustion.

In addition to the structured activity protocol, the dataset includes a set of annotated semi-structured activities, each with a duration of approximately two minutes, representative of common daily-life behaviors. These activities were: Standing while sweeping, Standing while folding towels, Standing while moving books, Standing while using a PC, Sitting while reading, Sitting while using a PC, Sitting while watching TV, Jogging, Walking with groceries, Walking with phone or book, Walking at usual speed, Zigzag walking, and Stair ascent and descent. For these activities, direct calorimetric measurements were not available, and metabolic intensity was therefore estimated using standardized values from the Compendium of Physical Activities. This combined design enables physiologically grounded validation where indirect calorimetry is available, while supporting broader evaluation under conditions representative of large-scale population cohorts.

Table XXX shows the equivalent MET used for each of them:

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D. Data Preprocessing

IMU data were recorded at a sampling frequency of 25 Hz with UNIX timestamps and synchronized prior to deployment using a laboratory computer. During the experimental sessions, the start and end times of each structured and semi-structured activity were documented in a laboratory log using the same clock, enabling subsequent identification, segmentation, and labeling of the corresponding signal segments within the continuous seven-day IMU recordings.

As no additional synchronization was performed after deployment, clock drift in the internal IMU timers resulted in temporal misalignment between the logged activity times and the sensor signals. This drift was corrected by visually identifying activity onset landmarks in the accelerometer data and applying a linear timestamp correction matching this with the actual onset of the activity registered in the laboratory log. Indirect calorimetry data obtained using the COSMED K5 system were recorded at 0.25 Hz and temporally aligned with the IMU signals through interpolation. The corrected and

synchronized signals were then segmented into 10-second windows with 50% overlap for subsequent analysis.

E. Feature Extraction

The window size used for time-series descriptor extraction was set to 10 seconds [14], as it represents a suitable compromise between effectively capturing activity intensity and accommodating variations or transitions in behavioral patterns.

For each signal window, a set of time- and frequency-domain descriptors was extracted based on features commonly used in the literature (see Section II – State of the Art), which have shown strong performance in physical activity assessment. Specifically, first-order statistical descriptors, including the minimum, maximum, median, 25th percentile, and 75th percentile, were computed for each accelerometer and angular velocity axis (x, y, z). Frequency-domain features were derived for each axis by estimating the power and frequency of the first and second harmonics, as well as the spectral entropy of the signal. In addition, peak-based descriptors, namely the number of peaks and the mean peak prominence, were computed per axis. Inter-axis coordination was characterized through Pearson correlation coefficients between the x–y, x–z, and y–z axis pairs, calculated separately for acceleration and angular velocity signals. Temporal dependency within movement patterns was further captured by computing the one-second lag autocorrelation of the acceleration vector magnitude. Finally, orientation information was summarized by calculating the window-wise mean values of the pitch, yaw, and roll angles.

F. Evaluated Models

To assess the predictive capability of different modeling strategies, several approaches from both traditional machine learning and deep learning were evaluated. The considered models were grouped according to the data representation used as input: feature-based models and time-series-based models. Under the feature-based approach, models received as input the set of engineered descriptors described in the previous section. The evaluated models in this category included Ridge Regression, Random Forest, XGBoost, and a feed-forward neural network. These methods enable the modeling of both linear and nonlinear relationships between the input variables and the target outcome, offering different trade-offs in terms of model complexity and generalization capability.

The second approach directly exploits the temporal structure of the data. In this case, observations were organized into fixed-length temporal windows, allowing the capture of temporal dependencies and dynamic patterns that may not be adequately represented through aggregated features. To analyze these temporal sequences, three deep learning architectures were evaluated: Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and Transformer models.

F. Hyperparameter tuning

To obtain suitable configurations for each of the evaluated models, a hyperparameter optimization stage was performed prior to final training. Owing to the different parameterizations

of the considered algorithms, this process was conducted independently for each model.

Hyperparameters were optimized using Bayesian optimization, whereby candidate configurations were iteratively evaluated based on their performance on a validation set. The root mean squared error (RMSE) was used as the selection criterion. For neural network-based models, early stopping was applied, halting training when the validation error did not improve for 10 consecutive epochs.

The searched hyperparameter ranges were defined according to common practices in the literature for each model class and are summarized in the Supplementary Material.

F. Model Training

Model training was performed using a cross-validation strategy. Specifically, a five-fold ($K = 5$) cross-validation scheme was applied, where each fold was iteratively used as the test set while the remaining four folds were used for model training. For each fold, the root mean squared error (RMSE) and the coefficient of determination (R^2) were computed. The mean and standard deviation of these metrics across the five folds were then reported to summarize model performance.

E. Evaluation Metrics

Model performance was evaluated using two complementary metrics: the Root Mean Squared Error (RMSE) and the coefficient of determination (R^2). Evaluation was conducted separately for each sensor configuration, wrist, thigh, and the combined wrist + thigh setup, allowing direct comparison across models and an assessment of how information from different sensor locations influences prediction accuracy and robustness.

To account for potential overfitting and obtain a more conservative performance estimate, an adjusted RMSE metric was computed:

$$RMSE_{adj} = RMSE_{test} + |RMSE_{test} - RMSE_{train}|$$

Likewise, an adjusted uncertainty estimate was calculated to reflect variability in both training and test performance:

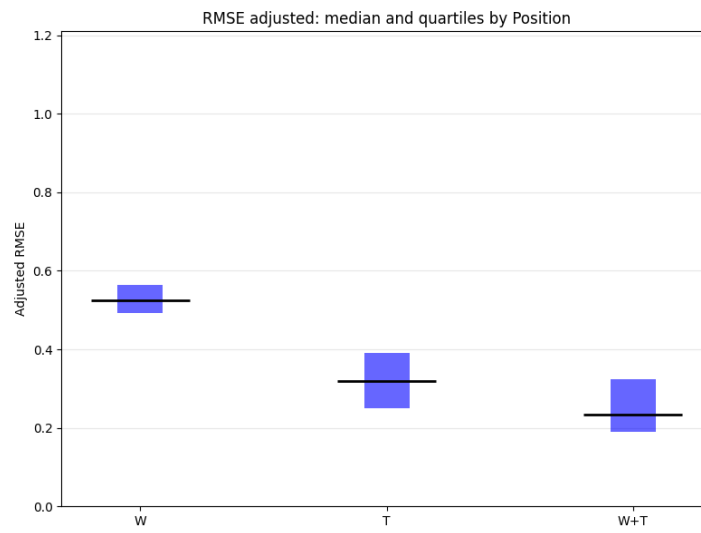
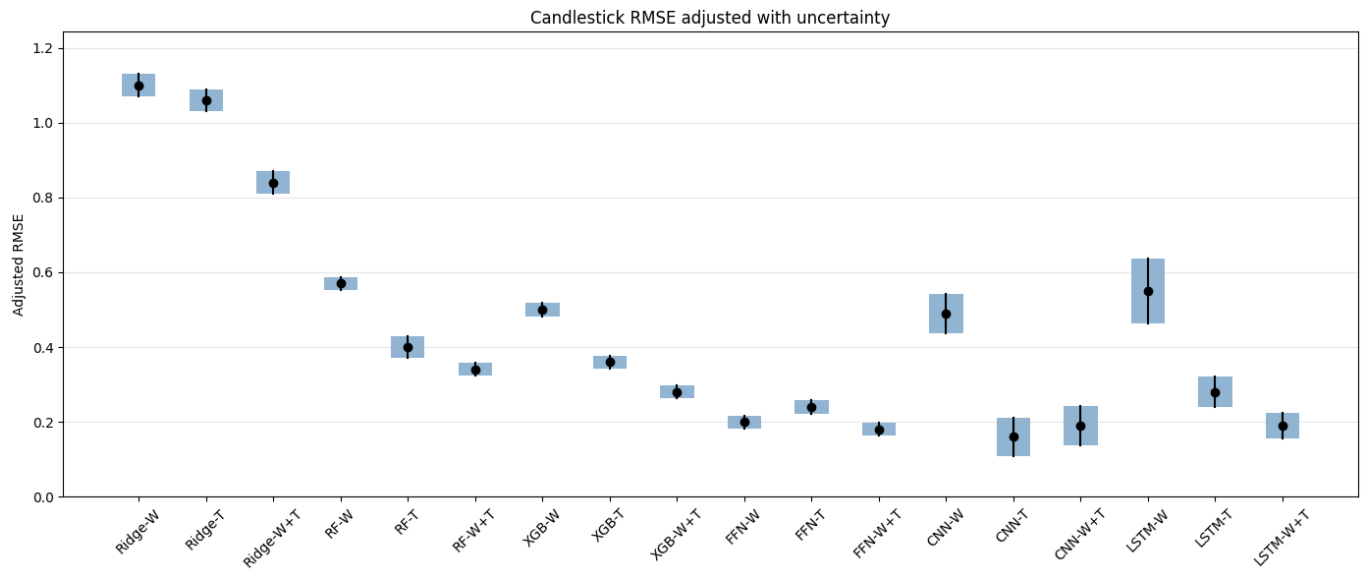
$$\sigma_{adj} = \sqrt{2\sigma_{test}^2 + \sigma_{train}^2}$$

This formulation penalizes models exhibiting large discrepancies between training and test errors, thereby favoring solutions that generalize more consistently across unseen data.

V. Resultados

A. Model Performance

Model	Pos	Train		Test	
		RMSE	R^2	RMSE	R^2
Ridge	W	1.08±0.01	0.74±0.01	1.09±0.02	0.74±0.01
	T	1.04±0.01	0.75±0.01	1.05±0.02	0.75±0.01
	W+T	0.82±0.01	0.80±0.01	0.83±0.02	0.80±0.01
RF	W	0.09±0.01	0.97±0.01	0.33±0.01	0.92±0.01
	T	0.06±0.01	0.98±0.01	0.23±0.02	0.94±0.01
	W+T	0.06±0.01	0.99±0.01	0.20±0.01	0.95±0.01
XGB	W	0.04±0.01	0.99±0.01	0.27±0.01	0.94±0.01
	T	0.04±0.01	0.99±0.01	0.20±0.01	0.95±0.01
	W+T	0.02±0.01	0.99±0.01	0.15±0.01	0.96±0.01
FFN	W	0.06±0.01	0.98±0.01	0.13±0.01	0.97±0.01
	T	0.04±0.01	0.99±0.01	0.14±0.01	0.97±0.01
	W+T	0.04±0.01	0.99±0.01	0.11±0.01	0.97±0.01
CNN	W	0.41±0.03	0.90±0.01	0.45±0.03	0.89±0.01
	T	0.10±0.03	0.98±0.01	0.13±0.03	0.97±0.01
	W+T	0.11±0.03	0.97±0.01	0.15±0.03	0.96±0.01
LSTM	W	0.39±0.05	0.91±0.01	0.47±0.05	0.88±0.01
	T	0.26±0.04	0.94±0.01	0.27±0.0	0.93±0.01
	W+T	0.09±0.02	0.98±0.01	0.14±0.02	0.97±0.01
Transf	W				
	T				
	W+T				



VI. Discussion

VII. Conclusion

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Supplementary Material. Hyperparameter Optimization

Hyperparameter optimization was conducted independently for each evaluated model to account for algorithm-specific parameterizations. The optimization process was performed prior to final model training using Bayesian optimization, whereby candidate hyperparameter configurations were iteratively evaluated based on their performance on a validation dataset. The root mean squared error (RMSE) was used as the objective function for all models. For neural network-based approaches, early stopping was applied, terminating training when the validation error did not improve for 10 consecutive epochs. Ridge Regression. For the Ridge Regression model, the L2 regularization parameter α was optimized. The search space was defined on a logarithmic scale within the range 10^{-4} to 10^0 . Random Forest. The following hyperparameters were optimized: number of trees (50–100), maximum tree depth (3–20), minimum number of samples required to split an internal node (2–10), and minimum number of samples required at a leaf node (1–10). XGBoost. Hyperparameter optimization included the number of trees (50–300), maximum tree depth (3–20), learning rate (0.01–0.3), subsampling ratio (0.5–1.0), column subsampling per tree (0.5–1.0), minimum gain required to perform a split (0–5), and L2 regularization strength (0–5). Feed-Forward Neural Network. The optimized hyperparameters included the number of hidden layers (1–3), number of neurons per layer (4–128), dropout rate (0.2–0.5), batch size (16, 64, or 128), learning rate (10^{-4} – 10^{-3} , logarithmic scale), and L2 regularization strength (10^{-3} – 10^{-2} , logarithmic scale). Convolutional Neural Network. The following hyperparameters were optimized: number of convolutional blocks (1–4), initial number of filters (16–64), filter growth factor per block (1 or 2), kernel size (3–7), dropout rate (0.2–0.5), L2 regularization strength (10^{-4} – 10^{-2} , logarithmic scale), learning rate (10^{-4} – 10^{-3} , logarithmic scale), and batch size (32, 64, or 128). Long Short-Term Memory Network. Optimization included the number of LSTM layers (1–3), number of units per layer (32–128), dropout rate (0.2–0.5), L2 regularization strength (10^{-4} – 10^{-2} , logarithmic scale), learning rate (10^{-4} – 10^{-3} , logarithmic scale), and batch size (32, 64, or 128). Transformer. Hyperparameters included the number of Transformer blocks (1–4), embedding dimensionality (32, 64, or 128), number of attention heads (2, 4, or 8), dimensionality of the internal feed-forward network (set between $2\times$ and $4\times$ the embedding dimension), dropout rate (0.2–0.5), learning rate (10^{-4} – 10^{-3} , logarithmic scale), and batch size (32, 64, or 128). Positional encoding was used to preserve temporal order information, and each Transformer block comprised multi-head self-attention, normalization, and feed-forward layers.

A. Hiperparámetros

Se presentan para cada modelo los hiperparámetros obtenidos mediante optimización bayesiana.

Ridge Regression

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
α	0.001	0.0001	0.00079

Random Forest

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
Estimators	81	77	92
Max depth	14	19	18
Min samples split	4	9	9
Min samples leaf	2	2	2

XGBoost

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
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Estimators	233	270	208
Max depth	17	19	11
Learning rate	0.08	0.03	0.06
Subsample	0.69	0.70	0.86
Colsample bytree	0.74	0.73	0.52
Gamma	1.04	1.51	0.59
Lambda	2.60	3.35	3.78

Feed-forward neural network

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
Layers	3	3	3
Neurons	114	128	79
Dropout	0.21	0.25	0.29
Learning rate	0.00018	0.00012	0.00027
Batch size	16	16	128
L2	0.0013	0.0015	0.0018

Convolutional neural network

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
Layers	4	4	4
Initial Filters	41	33	37
Filter growth	2	2	2
Kernel size	5	6	5
Dropout	0.35	0.20	0.22
L2	0.002	0.0001	0.0004
Learning rate	0.0007	0.0004	0.0004
Batch size	128	64	64

Long Short-Term Memory network

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
Layers	3	3	3
Neurons	46	76	79
Dropout	0.46	0.47	0.29
L2	0.0001	0.0009	0.002
Learning rate	0.0005	0.0001	0.0002
Batch size	32	128	128

Transformer

Hiperparámetro	Muñeca	Muslo	Muñeca+Muslo
Layers	3	3	4
d_embedding	64	64	64
N heads	4	8	4
d_Feed Forward	192	235	181
Dropout	0.16	0.34	0.25
Learning rate	0.0009	0.0002	0.0003
Batch size	128	32	64

